

Humanoid-Robot Actuators as a Manufacturing Business in India: Market Outlook to 2035, Copper Demand, Manufacturing Economics and a Go-to-Market Strategy

JX1 Humanoid Project

Abstract—Every humanoid robot needs 20–40 joint actuators, which make up about half of its bill of materials. This paper evaluates an Indian company that would manufacture such actuators. Combining bank forecasts, industry statistics and a bottom-up cost model, we find that the humanoid-actuator market grows from about US\$0.23 bn in 2025 to US\$8–14 bn in 2030 and US\$10–90 bn in 2035, and that India, which installs about 10,000 industrial robots a year, has no volume producer of robot-class actuators. Copper, often cited as the metal of the robot boom, is not a material driver: a humanoid holds about 6–8 kg, and humanoids remain below 0.1 % of world copper demand in 2030. We formalise the planning equations of an actuator plant (margin, break-even, learning curve, takt capacity, working capital, capital requirement and valuation), and apply them to a four-stage plan from a garage to a 60,000-unit-per-year plant, which needs about INR 105.9 crore (US\$11 M) of external capital and reaches about INR 126.0 crore of yearly revenue. We conclude that the business is viable if it targets buyers that value trust, service and customisation over price, earns revenue first as an integrator, and scales tooling only against signed orders. A US\$10 bn outcome would require about a quarter of the entire 2030 actuator market.

Index Terms—Humanoid robots, actuators, market analysis, copper demand, manufacturing economics, learning curve, venture capital, go-to-market, India.

I. INTRODUCTION

HUMANOID robots went from a few hundred units in 2023 to 13,000–20,000 in 2025, almost all built in China [2], [11]. Their cost is dominated by actuators, and Chinese actuator makers (RobStride, Encos, Unitree) already produce 50,000–500,000 joints a year at about CNY 1,000 each [13]. A companion paper [1] showed that an Indian builder can design and make RobStride-class actuators, but not more cheaply at small volume. This paper asks the business questions that follow:

- 1) How large is the actuator market, and how fast does it grow?

Manuscript prepared October 2026 as part of the open-source `thenar-jx1` repository (<https://github.com/nickthelegend/thenar-jx1>). The business model is `business/actuator_business_model.py`; source data with evidence labels are in `research/raw/`. Forecasts are scenarios, not predictions, and nothing here is investment advice.

TABLE I
HUMANOID FORECASTS

Source	2030	2035	Long run
BofA Institute, Mar. 2026 [2]	1.2 M/yr	10 M/yr	3 bn in use (2060)
Goldman Sachs, Sep. 2026 [†]	0.89 M	6.5 M; US\$138 bn	–
Morgan Stanley [†]	US\$28 bn	13 M in use	US\$5 trn (2050)
UBS [†]	–	US\$30–50 bn	300 M (2050)
Citi GPS, Nov. 2024 [3]	–	–	US\$7 trn (2050)

[†]From press coverage; original notes paywalled [11].

- 2) Does the robot boom make copper the key material, and does copper matter to an actuator maker?
- 3) Who makes actuators in India, and how are such plants built and financed?
- 4) What capital does an Indian actuator company need, what is it plausibly worth, and how should it enter the market?

II. MARKET OUTLOOK

A. Humanoids

Table I summarises published forecasts. Estimates for 2035 differ by about 5× and every bank raised its forecast in the last year. In BofA’s 2030 bill of materials, rotary and linear actuators are 51 % of a humanoid’s cost (dexterous hands a further 19 %) [2], so the actuator market follows as

$$M_{\text{act}} = N_{\text{robots}} \cdot C_{\text{BOM}} \cdot s_{\text{act}}, \quad (1)$$

giving about US\$0.23 bn in 2025 ($13\text{k} \times \text{US\$}35\text{k} \times 0.51$), US\$10.4 bn in 2030 ($1.2\text{M} \times \text{US\$}17\text{k} \times 0.51$) and US\$41–87 bn in 2035 depending on the source.

B. Industrial robots and India

More than 600,000 industrial robots were installed worldwide in 2025, 59 % of them in China, and IFR expects 806,000 in 2029 [4]. India installed about 9,000–10,500 (sixth in the world), which corresponds to about 55,000 servo joints a year, all imported [12]. Domestic humanoid demand to 2030 is likely in the thousands of units (Addverb targets 3,000 installations; MeitY announced a INR 500 crore humanoid fund in June 2026), so an Indian

TABLE II
COPPER INTENSITY AND HUMANOIDS' SHARE OF DEMAND
(ESTIMATED)

Item	Value
World refined demand 2025 / 2030 / 2040	28 / ≈ 32 / 42 Mt
Copper per battery EV / per MW data centre	83 kg / 20–40 t
Copper per humanoid	6–8 kg ($\approx$ US\$115)
Copper in one 120 N·m actuator	≈ 0.1 kg
Humanoid share of demand, 2030	0.02–0.08 %
Humanoid share of demand, 2035	0.14–0.33 %
Humanoid share with 1 bn robots in use (2040+)	≈ 4 %

actuator company needs export and non-humanoid volume from the start.

III. COPPER DEMAND

World refined-copper demand was about 28 Mt in 2025 and is projected at 42 Mt in 2040, with supply peaking near 33 Mt in 2030 and a shortfall of about 25 % by 2040 [5]. The share of demand attributable to humanoids is

$$s_{\text{Cu}} = \frac{N_{\text{robots}} m_{\text{Cu}}}{D_{\text{world}}}, \quad (2)$$

with $m_{\text{Cu}} \approx 6\text{--}8$ kg per humanoid (windings 2–7 kg, harness, boards, battery), consistent with CRU (4–8 kg) and Morgan Stanley (4.5–8.5 kg) estimates [11]. Table II gives the result: 0.02–0.08 % in 2030 and 0.14–0.33 % in 2035, compared with about 6 % for electric vehicles (83 kg each) and a growing share for AI data centres (20–40 t/MW). Copper prices are near record levels (about US\$14,500/t, +30 % year on year) for reasons unrelated to robots. For an actuator maker copper is a minor cost: a JXA-120 contains about 0.1 kg of winding wire, so a 30 % copper price rise adds about INR 75 per unit, whereas a magnet export restriction can halt production.

IV. SOFTWARE AND HARDWARE LABOUR TO 2030

Evidence to date shows AI compressing entry-level programming rather than software work as a whole: employment of US developers aged 22–25 fell by about 20 % after 2024 while older developers grew 6–12 % [6], young workers in AI-exposed occupations sit 19 % below trend through reduced hiring [7], yet the BLS projects software-developer employment to grow 15.8 % over 2024–34 while “computer programmers” decline 6 % [8]. Robots displace physical work slowly: one additional robot per 1,000 workers lowers the employment-to-population ratio by about 0.2 percentage points [9]. Actuator products are roughly half electromechanical and half software (field-oriented control, calibration, simulation models), so a founder combining software with manufacturing know-how holds a durable position.

V. THE INDIAN ACTUATOR LANDSCAPE

Table III lists Indian firms relevant to robot actuation. No Indian company produces RobStride-class actuators at volume and no Indian harmonic-reducer maker was

found; startups build quasi-direct-drive joints by hand in tens to hundreds, and others import. India does, however, have the adjacent capabilities at scale: drone motors (Vector Technics, 300,000 units/yr with in-house winding), EV traction and hub motors (Sona Comstar, Lucas TVS, Chara), BLDC fan motors (Atomberg, millions per year) and lamination stamping (Pitti Engineering, 90,000 t/yr). Relevant manufacturing clusters are Coimbatore (motors, winding), Rajkot (CNC, shafts), Pune–Chakan (gears), Bengaluru Peenya (aerospace machining), Hyderabad (drone motors, laminations, defence) and Chennai–Hosur (drives, EV motors).

VI. MANUFACTURING ARCHITECTURES AND PLANNING EQUATIONS

A. Architectures

Three architectures are observed. (A) *Design and assembly house*: design, winding, assembly, calibration and test in-house; all parts from job shops; 100–3,000 units/yr; equipment about INR 15.7 lakh (the kits in [1]). (B) *Partly integrated*: adds automated needle winding, magnet bonding, gear finishing and end-of-line test; 3,000–30,000 units/yr; about INR 1–3 crore for 10,000/yr. (C) *Volume plant*: adds stamping, die casting, gear hobbing or powder metallurgy and automated lines; 50,000–1,000,000 units/yr; INR 20–100+ crore (a single Chinese joint-module line is reported at CNY 110 M [13]). The process flow is: incoming inspection, lamination stack, insulation, needle winding, termination, hipot test, impregnation, rotor bonding and balancing, gear stage, cast-and-machined housing, final assembly, firmware and encoder calibration, end-of-line test (back-EMF, K_t , cogging, thermal, backlash), burn-in and packing. Chinese benchmarks are 90 s per joint and over 96 % first-pass yield.

B. Planning equations

With price P , unit cost C , variable cost V and fixed cost F per year, gross margin and break-even volume are

$$\text{GM} = \frac{P - C}{P}, \quad N^* = \frac{F}{P - V}. \quad (3)$$

Unit cost falls with cumulative volume according to Wright’s law [10],

$$C(N) = C_1 N^{\log_2 r}, \quad (4)$$

where r is the cost ratio per doubling; our cost model gives $r \approx 0.83$ between 25 and 1,000 units per year. Line capacity follows from the takt time t_{takt} and overall equipment effectiveness (OEE),

$$N_{\text{line}} = \frac{t_{\text{avail}} \text{OEE}}{t_{\text{takt}}}, \quad (5)$$

so one line at a 90 s takt, two shifts over 300 days and OEE of 0.65 makes about 124,800 actuators a year: capacity is

TABLE III
INDIAN ACTUATOR AND MOTOR LANDSCAPE (PRESS AND COMPANY SOURCES; MOST FUNDING FIGURES UNVERIFIED) [12]

Company	Product	Manufacturing approach	Scale / funding
xTerra Robotics (Kanpur/Bengaluru)	QDD A2: 12 N·m, 550 g, INR 94,400	parts from aerospace job shops, in-house assembly	≈15 staff, ≈INR 1 crore revenue, grants
Bhairav Robotics (Kakinada)	“Prabal” QDD, quadrupeds, UGVs	in-house	Zen Technologies acquired 45 % for ≈INR 4 crore
Perceptyne (Hyderabad)	dual-arm semi-humanoid, own actuators	in-house	US\$3 M seed (Endiya, Yali), 2024
Addverb (Noida; Reliance)	wheeled humanoid (≈US\$45k), BLDC + planetary joints	undisclosed	FY25 revenue INR 800 crore; INR 75 crore plant for 60k robots/yr
General Autonomy (Bengaluru)	Atom 01 humanoid, quadruped	imports actuators	seed INR 32 crore at INR 280 crore valuation
Vector Technics (Hyderabad)	drone propulsion motors	own winding and machining, 300k/yr	Zen paid ≈INR 25 crore for 51 %
Sona Comstar / Lucas TVS / Chara	EV traction and hub motors	automated winding and stamping	INR 99.7 crore for 200k motors/yr (Sona)
Atomberg; Pitti Engineering	BLDC fans; laminations	mass stamping and winding	6.6 M units/yr; 90k t/yr

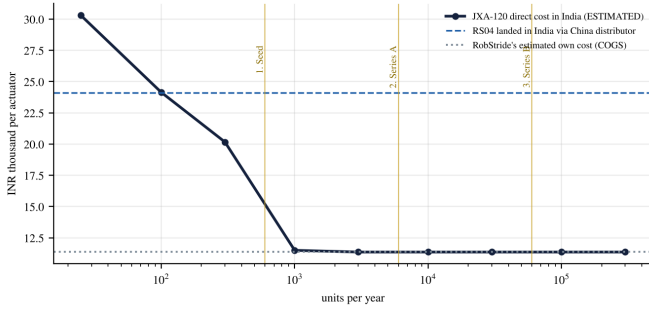


Fig. 1. JXA-120 direct cost versus yearly volume (estimated). Cost falls steeply up to about 1,000 units/yr as job shops are replaced by own tooling, then is floored at the Chinese manufacturer’s estimated cost, because magnets, bearings and chips cost everyone the same.

not the constraint; demand, cost and quality are. Working capital and the capital to raise for a stage are

$$WC = \frac{COGS}{365} (d_{inv} + d_{rec} - d_{pay}), \quad (6)$$

$$K = CAPEX + \sum \text{operating losses} + WC + 0.5 OPEX, \quad (7)$$

with $d_{inv} = 120$, $d_{rec} = 60$ and $d_{pay} = 30$ days, inventory being long because magnets and bearings are imported. Finally, valuation follows revenue:

$$MC = m \cdot R, \quad R = N_{\text{robots}} n_{\text{act}} s P, \quad (8)$$

with EV/sales multiple m , n_{act} actuators per robot and market share s .

VII. CAPITAL REQUIREMENT

Table IV applies the equations to a four-stage plan, with units expressed as 120 N·m actuator equivalents, prices falling as volume rises and cost floored at RobStride’s estimated cost (INR 11,357). Total external capital is about INR 105.9 crore (US\$11 M), dominated by the Series B plant (Fig. 2).

Three observations follow. First, Stage 1 is close to break-even (about 692 units/yr) only because laboratories

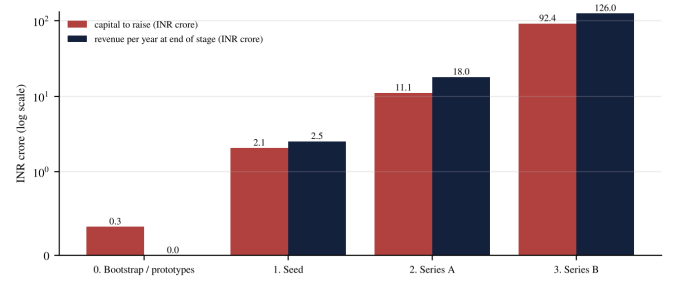


Fig. 2. Capital to raise per stage versus yearly revenue reached at the end of the stage (INR crore, logarithmic scale).

and defence buyers pay about INR 42,000 for a 120 N·m joint, still less than half the Indian retail price of a CubeMars AK10-9. Second, the critical risk is Stage 3: at INR 21,000 per unit the gross margin is 46 %, and if Chinese price cuts force INR 15,000 it falls to 24 %; volume tooling should therefore be built only against signed contracts. Third, at the end of Stage 3 the company earns about INR 126.0 crore (US\$13 M) a year, worth about US\$26–53 M at 2–4× sales.

Table V lists the capital sources found. Indian robotics start-ups raised only US\$52.9 M in 2025, about 0.5 % of Indian technology venture funding (the US raised US\$5.6 bn) [12], so grants, defence programmes, customer pre-payments and strategic investors matter more than in software.

VIII. VALUATION

Listed component makers trade at 0.3–3.5× sales when mature (Nabtesco 1.9×, Inovance 2.9×, Schaeffler 0.25×), 4.5–10× when priced as humanoid suppliers (Sanhua 4.5×, Harmonic Drive 10×), and 75–88× for pure plays such as Leaderdrive and Unitree, whose shares opened 629 % above the IPO price in August 2026 and have since fallen about 59 % from that open [11]. One per cent of the 2030 actuator market is about US\$104 M of revenue, worth about US\$416 M at 4× sales. Table VI shows what a US\$10 bn actuator company would require.

TABLE IV
STAGED PLAN (ESTIMATED; AMOUNTS IN INR CRORE UNLESS NOTED)

Stage	Months	Units/yr	Price [INR]	Unit cost [INR]	GM	Revenue/yr	Opex/yr	EBITDA/yr	Capex	Raise
0. Bootstrap / prototypes	9	12	–	30,292	–	0.0	0.1	-0.2	0.1	0.3
1. Seed: first customers	18	600	42,000	18,000	57 %	2.5	1.7	-0.2	0.5	2.1
2. Series A: product line	24	6,000	30,000	11,357	62 %	18.0	8.6	2.6	4.0	11.1
3. Series B: volume plant	24	60,000	21,000	11,357	46 %	126.0	38.8	19.1	45.0	92.4

TABLE V
CAPITAL SOURCES FOR AN INDIAN ACTUATOR COMPANY [12]

Source	Size
Bootstrap (garage kit + prototypes)	INR 10–35 lakh
Startup India Seed Fund (via incubator)	≤INR 20 lakh grant + 50 lakh debt
Incubators: ARTPARK, IIT hubs (DST NM-ICPS)	grants, labs
State grants: Karnataka ELEVATE, TANSEED	≤INR 50 / 10–15 lakh
Defence: iDEX / ADITI	≤INR 1.5 / 25 crore
MeitY humanoid fund (announced)	INR 500 crore total
Seed VC (Yali, Speciale, Endiya, pi, Kalaari, Blume)	US\$2–3.5 M
Series A/B VC	US\$5–10 M / 20–30 M
Strategic (Zen, Reliance, Zoho)	stake or acquisition
SIDBI venture debt; credit guarantee	INR 10–15 / ≤20 crore
ECMS component scheme	25 % capex support (large)

TABLE VI
REQUIREMENTS FOR A US\$10 BN ACTUATOR COMPANY (ESTIMATED)

EV/sales	Revenue [US\$ bn]	Actuators/yr at US\$250 [M]	Share of actuator market		
			2030	2035 low	2035 high
2×	5.0	20	48 %	12.2 %	5.7 %
4×	2.5	10	24 %	6.1 %	2.9 %
10×	1.0	4	10 %	2.4 %	1.1 %

IX. GO-TO-MARKET STRATEGY

A. Positioning

An Indian entrant cannot win on price against CNY 1,000 joints. It can win on attributes some buyers value more: *trust and compliance* (defence, government laboratories and Western buyers avoiding Chinese components), *local support* (Indian stock, GST invoices, two-day replacement), *customisation* (ratios, shafts, IP rating, 72 V windings, and the 360 N·m class nobody sells cheaply) and *software* (a RobStride-compatible CAN protocol, ROS 2 drivers and accurate simulation models for reinforcement learning).

B. Beachhead customers

In order: (1) university and research laboratories, which pay INR 34,000–1,05,000 per imported joint today; (2) Indian robotics start-ups building quadrupeds, humanoids, cobots and exoskeletons, all of which import joints; (3) defence programmes (iDEX challenges, DRDO laboratories, integrators) where a non-Chinese supply chain earns a

TABLE VII
GO-TO-MARKET SEQUENCE

Months	Offer	Milestone
0–6	integration and resale of imported joints with Indian software, stock and support	10 paying customers; JXA-120 v1 on the dynamometer
6–18	JXA-120 and JXA-40, hand-assembled, to laboratories and start-ups; first iDEX bid	600/yr run-rate; 200 h life data; seed round (INR 2.1 crore)
18–36	JXA-360 and 72 V line; cast housings; own driver board; export	6,000/yr; ISO 9001; Series A (INR 11.1 crore)
36+	volume contracts with humanoid makers; automated winding and test	60,000/yr; Series B or strategic partner (INR 92.4 crore)

premium; (4) export customers seeking a second source; and (5) later, replacement of the roughly 55,000 industrial servo joints India imports each year.

C. Sequencing

Table VII gives the sequence. The first product is *integration*: imported joints sold with Indian-built interface boards, ROS 2 drivers, simulation models, local stock and warranty, which earns revenue and customer relationships while the in-house actuator is developed. Kill criteria are set in advance: if fewer than five customers re-order within 18 months, or the 120 N·m unit cannot reach INR 20,000 at 600 units/yr, the company should remain an integrator and software firm rather than a manufacturer.

X. RISKS

The principal risks are: Chinese price cuts (RobStride targets 300,000 units in 2026), mitigated by never competing on price alone; rare-earth magnet export licences (the wider Chinese rules are suspended only until 10 November 2026), mitigated by stockpiling magnet sets, qualifying dysprosium-free N48H grades and following India’s INR 7,280 crore magnet scheme; a humanoid hype cycle, mitigated by selling to non-humanoid robots and raising on revenue; field failures, mitigated by end-of-line testing of every unit and life testing; and thin Indian hardware venture capital, mitigated by grants, defence programmes and strategic investors.

XI. CONCLUSION

An actuator company (“selling shovels”) is a sounder venture for an Indian robotics founder than another humanoid start-up: demand is real, growing about 40-fold

from 2025 to 2030, and every Indian robot maker imports its joints today. It is, however, a manufacturing business with 25–60 % gross margins at best, competing with Chinese makers that are already at volume. The recommended path is to earn revenue first as an integrator, sell in-house actuators to laboratories, start-ups and defence buyers on trust, service and customisation, raise capital in small steps (about INR 2.1 crore at seed and INR 11.1 crore at Series A), and build volume tooling only against signed orders. Copper will likely remain expensive because of grids, electric vehicles and data centres, but it is not the lever for an actuator maker; magnet supply and quality are.

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